

Degradation-Trend-Aware Deep Neural Network With Attention Mechanism for Bearing Remaining Useful Life Prediction

Yongkang Liu , Student Member, IEEE, Donghui Pan , Haifeng Zhang , and Kai Zhong 

Abstract—Remaining useful life (RUL) prediction of bearings has extraordinary significance for prognostics and health management (PHM) of rotating machinery. RUL prediction approaches based on deep learning have been dedicated to finding a nonlinear mapping relationship between nonstationary monitoring data and RUL. However, most existing approaches pay little attention to the degradation trend of diverse health stages of bearing and lack the discriminative power of crucial degradation features, resulting in the loss of some important information associated with RUL. To address this challenge, this article proposes a novel RUL prediction framework based on degradation-trend-aware deep neural network with attention mechanism (DTADAN). First the multidirection features with evident degradation trend are extracted via the analysis of bearing vibration signal from both time-domain and time-frequency domain. Next, the deep neural network architecture with attention mechanism is utilized to adaptively learn the critical degradation features beneficial for RUL prediction. Distinct from the existing approaches, the proposed framework is able to dynamically extract key degradation features of the bearing including degradation trend information and effectively fuse multidirection information to improve RUL prediction accuracy. The performance of the proposed approach is evaluated via case studies on XJTU-SY bearing dataset and PRONOSTIA bearing dataset. Compared with other state-of-the-art approaches, the proposed framework has better predictive accuracy and robustness. Additionally, interpretable analysis is provided to reveal the process of model learning and data characteristics, and the analysis results are helpful in guiding model learning.

Impact Statement—This article aims to address the practical issue of RUL prediction of the bearing widely used in industrial equipment, which can help avoid unscheduled downtime, reduce

the risk of equipment failures and improve reliability. Traditional deep learning based prediction approaches do not incorporate degradation trend information and lack interpretability. To overcome this limitation, an innovative RUL prediction framework has been proposed for the unit operated under a single working condition, which has three main advantages: 1) efficient extraction of crucial degradation features through the appropriate use of attention mechanism; 2) utilization of degradation trend information from both the network structure and model training perspectives; and 3) detailed analysis of the interpretability for the entire model learning process. The proposed prediction framework achieves both interpretable RUL prediction and superior accuracy when compared to some current advanced approaches. The presented approach and interpretability analysis hold significant importance for intelligent prognostics and practical industrial applications in PHM field.

Index Terms—Attention mechanism, bearings, deep learning, degradation trend, interpretability, remaining useful life.

I. INTRODUCTION

PROGNOSTICS and health management (PHM) of rotating machinery aims to improve the equipment reliability and reduce maintenance costs, which is a key part of achieving smart manufacturing [1]. The remaining useful life (RUL) prediction is an extremely critical part of PHM and a precursor to maintenance strategy development. Bearings are a wearable core component of rotating machinery, such as aero engines, electric motors, and wind turbines [2], [3], [4]. The failure of the bearing can result in unpredictable consequences and substantial economic losses. However, due to the complex and variable operating conditions of bearings, the degradation process of different bearings shows heterogeneity, and the failure mode is difficult to be identified. Therefore, it is still a great challenge to make accurate RUL predictions of bearings.

As a general rule, RUL prediction approaches can be divided into model-based and data-driven approaches. The model-based approaches attempt to establish specific physical models of degradation processes. However, owing to the lack of a priori knowledge of the complex operating processes of rotating machinery, it is often extremely difficult to understand the physics of damage, which leads to the difficulty to construct a well-defined physical model. With the development of sensor technologies, massive condition monitoring data can be available so the various data-driven approaches have been extensively

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researched. In particular, with the increasing maturity of deep learning (DL) approaches, it has shown extremely fast learning speed and excellent application in a considerable number of fields. Therefore, the data-driven approaches are widely proposed to predict the RUL of bearings, which require less prior knowledge and exhibit better generalization ability.

The data-driven solutions for predicting the RUL of bearings are becoming increasingly sophisticated. Artificial intelligence (AI) approaches have been widely applied to perform the RUL prediction of mechanical equipment [5]. Recently, the shallow machine learning models have been utilized for RUL prediction of the bearing, including support vector regression (SVR) [6] and random forests (RF) [7] which are both AI technologies based on statistical learning. Additionally, some shallow neural network structures like multilayer perceptron (MLP) [8] have also been utilized for RUL prediction. However, the shallow machine learning model has its limitations in terms of model size, it usually shows poor performance in handling complex, coupled, nonlinear, and nonstationary issues in mechanical equipment monitoring data. As a result, RUL prediction methods are increasingly relying on complex DL structures to enhance the deep feature mining capability of models. DL-based prediction approaches have the ability to extract the nonlinear degradation patterns related to RUL prediction of the bearing, such as deep convolutional neural networks (DCNN) [9], gated recurrent unit network (GRU) [10], long short-term memory (LSTM) [11], [12], and deep belief networks (DBN) [13].

Many scholars utilize unsupervised learning networks, such as autoencoder (AE) [14], etc., to fuse the degradation feature to obtain health index (HI) and combine them with other RUL prediction methods. Hu et al. [13] proposed a DBN-LLE model that fuses manually crafted features in both time and frequency domains into HI, and then combines the Wiener process to predict the remaining useful life of bearings. Qin et al. [14] presented a variational autoencoder (VAE) model to construct HI with degradation-trend constraints, and then employed a macroscopic-microscopic-attention-based LSTM to perform the RUL prediction. Wu et al. [15] improved the AE structure for the characteristics of the temporal data and used LSTM as the encoder-decoder to greatly assist RUL prediction. These methods all achieve the prediction of RUL from bearing condition monitoring data, but different bearings often obtain different HI and the failure threshold also shows great individual differences [15]. In addition, virtual HI usually lacks physical meaning, which results in a lack of interpretability for prediction models. Therefore, many research works utilize labeled historical data as supervised training input and focus on developing end-to-end prediction methods directly from data to RUL. Wang et al. [16] proposed a RUL prediction framework based on deep separable convolutional network by combining separable convolutional layers, the concept of ResNet and squeeze and excitation unit. Wan et al. [17] utilized a convolutional LSTM to predict the RUL of bearings. Que et al. [18] proposed a GRU-based RUL prediction method and developed an attention mechanism based on dynamic time warping to improve prediction results. Wei et al. [19] uses

Siamese LSTM to classify the degradation stages of bearings before predicting their RUL. Additionally, a self-adaptive graph convolutional network (SAGCN) is introduced for accurate RUL prediction.

In order to compensate for the shortcomings of signal processing methods in existing research and better reveal the nonstationarity of vibration signals after degradation, many RUL estimation methods for bearings have introduced time-frequency analysis technology. This technology decouples detailed information in vibration signals and generates 2-D grey-scale energy maps known as time-frequency representations (TFRs). Some research efforts use TFRs as raw features and design various network structures to extract degradation-related information from them. Zhu et al. [9] employ a multi-scale DCNN to extract deep feature representations from TFRs at each time step, but this work ignored temporal information. Huang et al. [20] improved the RUL prediction results by supplementing manually crafted features that can be extracted in the time-domain. However, this work also only utilizes features at each moment and does not take into account temporal information. Zhuang et al. [21] introduced the transfer learning theory to address the challenges of bearing RUL prediction under multiple operating conditions, thereby enhancing the practical applicability of RUL prediction method. This approach incorporates temporal information and delivers outstanding predictive performance. However, when establishing degradation models, it is still necessary to pay further attention to the degradation feature extraction, utilization, and interpretability analysis of temporal information related to RUL prediction.

Most of the existing DL methods have been devoted to improving the network architecture used for RUL prediction, while ignoring the information representation itself. Time domain and frequency domain features alone are insufficient to fully characterize the intrinsic information of a signal, as they contain different degradation information. To improve the richness of information related to RUL prediction, data containing multidirection information can be constructed by utilizing the complementarity of time domain and time-frequency domain features. Utilizing the abundant information in the vibration signal through multiangle representation can improve the accuracy of RUL prediction [22]. In this case, constructing data samples with multiangle representations of the original signal can reduce information loss. Subsequently, effective extraction and fusion of deep degradation features from different perspectives is also a major challenge for RUL prediction.

As a typical mechanical component, bearings undergo continuous degradation from service to failure. The degradation phenomena accumulate continuously during the working stage of bearings, such as wear caused by poor lubrication and cracks caused by material fatigue [23]. When the deterioration begins to occur, different types of damage begin to accumulate at different rates, ultimately leading to the failure of bearings. The degradation damage of bearings ranges from nothingness to severity, reflecting trends that show significant differences due to various failure modes. Furthermore, individual differences among diverse bearings [5] and the mixture of different failure modes [24] result in varying degradation trends. At this point,

the degradation trend not only reflects changes in bearing health status but also implies unique properties possessed by each bearing. On this basis, the perceived degradation trend information is quite important for modeling the degradation process of the bearing and degradation trend awareness can mine out such key features that are beneficial to RUL prediction.

However, traditional RUL prediction models have inherent limitations in extracting crucial information. Their ability to represent important features is insufficient, especially when dealing with multidirectional data. Multidirectional feature data samples are created by representing raw vibration signals from multiple angles, which contain degradation information, redundant information and noise. Furthermore, the weights of different features and correlations between various pieces of information also change dynamically with the degradation of bearings. Unfortunately, it is challenging for these structures to perceive the degradation trend of bearings and capture detailed information from multidirectional feature data.

Various attention mechanisms have been successfully introduced into DL-based RUL prediction field to enhance the ability of networks to represent key degradation features [25]. Attention mechanisms in RUL prediction models are generally utilized to assign different weights to various segments of temporal sequences, which depict the relationships between different features. And this enhances the focus of the RUL prediction model on crucial features during the training phase [26]. These attention mechanisms can be formally divided into three categories: channel attention (CA) [27], [28], [29], spatial attention [30], and temporal attention [31]. The weights of these attentions are usually derived through squeeze-and-excitation network (SENet) paradigm [32] or self-attention mechanism [33], [34], which mainly achieve the enhancement of important information and suppression of irrelevant information or noise. For instance, Wang et al. [32] incorporated channel and temporal attention blocks to improve the feature representation of multisensor prediction models. Kong et al. [35] introduced graph attention (GAT) to capture spatial dependencies in multisensor data.

Although the application of the attention mechanism in prediction models on multisensor data can have a certain role in vibration data, the unique characteristics of vibration data require a redesign of the attention structure for bearing RUL prediction. The existing research efforts for multisensor devices mainly focus on learning temporal information and spatial correlation between multidimensional condition monitoring data simultaneously. Nevertheless, the dynamic correlations between different channels should be given special consideration for bearing vibration data. Additionally, the degradation information of bearings has varying priorities across each channel with specific key information contained within each feature, thus important information should be strengthened. Li et al. [34] improve LSTM by a self-attention mechanism that balances important feature extraction and preservation of temporal information. However, the embedded self-attention model lacks the focus on detailed information and interpretability. Xu et al. [30] apply residual structures to integrate more

information and spatial attention to enhance important features, but without the capability of extracting features in the temporal direction. Jiang et al. [31] propose a dual attention mechanism to enhance channel and temporal feature representation ability in shallow structures. However, the method lacks an attention design for deep structures. In particular, Fan et al. [36] propose a trend attention mechanism that emphasizes the decisive role of degradation trend information in prediction. However, this attention mechanism only considers a single type of information and ignores the enhancement of key detailed information by position encoding. Existing approaches often fail to extract degradation features efficiently from bearing vibration signals and lack interpretability, which only takes specific degradation information into account and without hierarchical processing of features.

To address these aforementioned challenges, this article proposes a RUL prediction framework of bearing based on degradation-trend-aware deep neural network with attention mechanism (DTADAN), which has the ability to perceive and fuse the RUL-related degradation feature information in multidirection representation. The main contributions of this article are as follows.

- 1) A RUL prediction framework relying on attention mechanism, i.e., DCNN with channel-spatial attention (CSA-DCNN) and LSTM with CA, is used to adaptively extract and fuse degradation features in multidirection representations. The proposed framework incorporates statistical indicators with explicit physical meaning and information of TFRs. The proposed method not only shows advantages in terms of prediction accuracy, but also provides interpretable analyses of model learning process and data characteristics from two perspectives.
- 2) The presented framework is able to effectively perceive the degradation trend information by means of attention mechanisms, which supervise the entire process of learning degradation trends with a well-designed configuration to overcome the insufficient representation of degradation trend features caused by weak attention embedding. Furthermore, the feature learning process has the capability of interpretation due to the intuitive design and specific meaning of the model structure.
- 3) The channel and spatial attention mechanisms are constructed with the help of SE paradigm so that the deep network architecture can adaptively identify RUL-related degradation features. This key feature representation capability significantly improves the problem of noise mixing and coupling of different types of information in high-dimensional multidirection feature data.

The remainder of the article is organized as follows. Section II briefly reviews some of the key basic theories. Section III gives the problem formulation and the details of the proposed framework. In Section IV, the prediction performance of the proposed approach is demonstrated by comparisons with some advanced approaches, and then the interpretability of the proposed approach is presented in detail. Finally, Section V summarizes the conclusions of this article and presents the potential of future work.

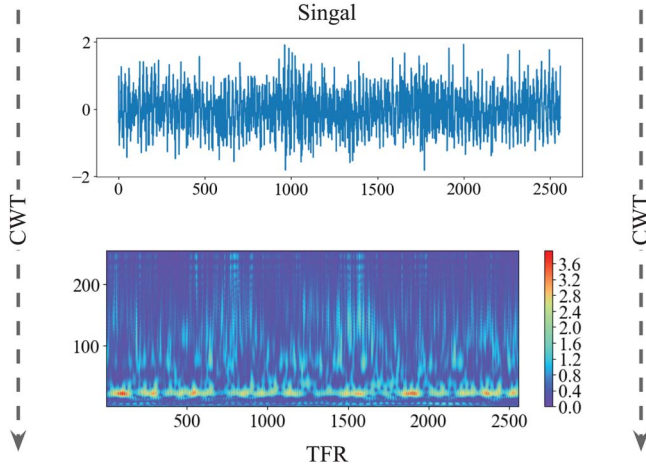


Fig. 1. Original signal and TFR.

II. BASIC THEORY

A. Continuous Wavelet Transform (CWT)

CWT is one of the time-frequency analysis methods. Usually, a standard wavelet function with a frequency-dependent scaling factor τ and a time-dependent translation factor s is chosen as the mother wavelet $\psi_{\tau,s}(t) \in L^2(R)$ in the first step. Then, the acquired raw signal is decomposed by translating and stretching the mother wavelet function. The resulting two-dimensional (2-D) matrix consisting of a series of stretching and translation coefficients is denoted as $H(\tau, s)$ (usually an energy map), and $H(\tau, s)$ is the TFR of the original signal (Fig. 1). The formula for calculating each TFR is shown below

$$\begin{aligned} H(\tau, s) &= \text{Dec}_{\text{CWT}} \langle \text{Signal}(t), \psi_{\tau,s} \rangle \\ &= \int_{-\infty}^{+\infty} \text{Signal}(t) \bar{\psi}_{\tau,s}(t) dt \\ &= \frac{1}{\sqrt{s}} \int_{-\infty}^{+\infty} \text{Signal}(t) \bar{\psi} \left(\frac{t-\tau}{s} \right) dt \end{aligned} \quad (1)$$

where $\text{Signal}(t)$ is the original one-dimensional (1-D) vibration signal collected at each moment, and $\bar{\psi}_{\tau,s}(t)$ is complex conjugate of $\psi_{\tau,s}(t)$. Since there is no classical or universal method for selecting a mother wavelet, the Morlet wavelet function is chosen as the mother wavelet function $\psi_{\tau,s}(t)$ because it resembles the impulse signal of a bearing [37].

B. Channel and Spatial Attention Model

CNN has a powerful hierarchical information extraction capability. Multiple convolutional kernels are always selected in CNN for extracting different features separately, so that each channel in the output features can be considered as a hierarchical representation of different types of information. Each layered representation is a 2-D feature distributions map containing specific details of each type of information.

To accommodate the characteristics of the output feature, a channel and spatial-based attention model is proposed. The dual attention mechanism is designed to evaluate which types

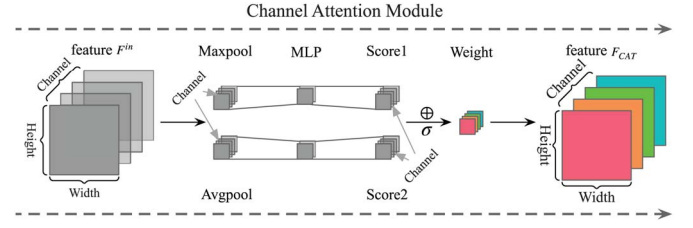


Fig. 2. Structure of the channel attention module.

of information and which details of them are beneficial for task [38]. The channel and spatial attention model can be expressed by the following formula

$$\begin{aligned} F_{\text{CAT}} &= \text{repeat} [\text{CAT}(F^{\text{in}})] \otimes F^{\text{in}} \\ F_{\text{SAT}} &= \text{repeat} [\text{SAT}(F^{\text{in}})] \otimes F^{\text{in}} \end{aligned} \quad (2)$$

where $F^{\text{in}} \in \mathbb{R}^{C \times H \times W}$ is the input feature, 1-D CA weight $\text{CAT}(\cdot) \in \mathbb{R}^{C \times 1 \times 1}$ and 2-D spatial attention weight $\text{SAT}(\cdot) \in \mathbb{R}^{1 \times H \times W}$, $\otimes$ indicates the Hadamard product operation. The details of each attention module are described below.

1) *Channel Attention Model*: As shown in Fig. 2. The maximum pooling and average pooling operations are performed for each channel of the input features, and two 1-D vectors with lengths equal to the number of channels are obtained as two-channel information representations. Then, an MLP with a squeeze layer and an excitation layer is introduced to derive the scores of different channels, and the channel weight vector is obtained after activation. Finally, the weights are multiplied elementwise into the input features to ensure that the dimensions of the features remain unchanged. The specific details are as follows:

$$\begin{aligned} \text{CAT}(F^{\text{in}}) &= \sigma [\text{MLP}(\text{avg-pooling}(F^{\text{in}})) \\ &\quad + \text{MLP}(\text{max-pooling}(F^{\text{in}}))] \\ &= \sigma [W_{\text{MLP}}^e (W_{\text{MLP}}^s (F_{\text{avg}}^c)) \\ &\quad + W_{\text{MLP}}^e (W_{\text{MLP}}^s (F_{\text{max}}^c))] \end{aligned} \quad (3)$$

where σ indicates the nonlinear activation function, i.e., “sigmoid” function, $W_{\text{MLP}}^s \in \mathbb{R}^{C/r \times C}$ and $W_{\text{MLP}}^e \in \mathbb{R}^{C \times C/r}$ are learnable parameters of squeeze and excitation layers in MLP. The ReLU activation function is utilized in MLP.

2) *Spatial Attention Model*: Unlike the CA model, maximum pooling and average pooling operate along the channel direction. The two obtained representations of location information are then considered as two channels input into a standard 2-D convolutional layer and fused to obtain a score for the spatial features. Then the position weights are obtained by activation. Finally, the Hadamard product of the weights and the input features are taken as the output, and the dimensionality is guaranteed to be constant. With specific reference to Fig. 3, the calculation formula is as follows:

$$\begin{aligned} \text{SAT}(F_{\text{in}}) &= \sigma [\text{Conv}^{7\text{CSA-DCNN}} ([\text{avg-pooling}(F^{\text{in}}) \\ &\quad \parallel \text{max-pooling}(F^{\text{in}})])] \\ &= \sigma [\text{Conv}^7 ([F_{\text{avg}}^s \parallel F_{\text{max}}^s])] \end{aligned} \quad (4)$$

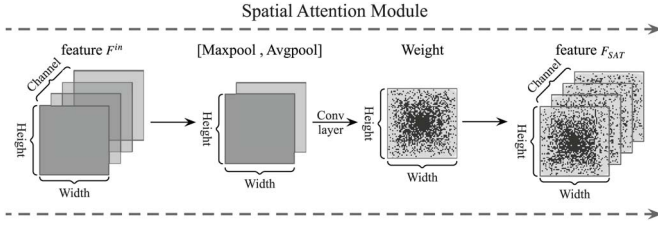


Fig. 3. Structure of the spatial attention module.

where σ denotes the sigmoid nonlinear activation function, and Conv^7 represents a convolution layer, and the kernel size is 7×7 .

III. THE PROPOSED RUL PREDICTION FRAMEWORK

In order to make full use of the information contained in the vibration signal and more accurately characterize the degradation status of the bearing, DTADAN is proposed for processing multidirection representations containing time domain features and TFRs, as shown in Fig. 4. The proposed RUL prediction framework consists of three main parts. First, multidirection features are constructed through the analysis of bearing vibration signals. Then, with the introduction of attention mechanism, the RUL-related degradation information is adaptively extracted from multidirection features. Specifically, CSA-DCNN and CA-LSTM are used to extract the significant degradation features related to bearing RUL prediction from the time-domain and time-frequency domain, respectively. Finally, the RUL prediction results are obtained by fusing embedding representation based on two fully connected layers.

A. Data Processing and Problem Statement

The new bearing is considered to be intact, but the components need to be adjusted slightly to break in with each other (wear phase) and then enter a stable working condition (healthy condition). During this time, it is very difficult to achieve an accurate prediction of RUL due to the stability monitoring signals. To determine the occurrence of damage and identify the first prediction time (FPT) position, as shown in Fig. 5, the method proposed by Li et al. [39] is introduced, which uses the 3σ criterion for the kurtosis index to determine the FPT.

When the damage of bearing occurs due to overload, lubrication degradation, etc., periodic impulse forces will be generated. These transient forces will increase the vibration level of the monitoring signal and will cause the bearing element to vibrate at its resonant frequency [40]. Finally, the monitoring vibration signal of the bearing shows a nonstationary character. Since CWT analysis can map a nonstationary vibration signal into a set of wavelet coefficients using the scaling and shifting of the mother wavelet. These coefficients, denoted as TFRs, can provide high time-frequency resolution and reveal the fine details in the monitoring vibration signal.

In addition, statistical indicators in the time-domain are also used as input data for the network in one direction. This is because these manually crafted features in the time-domain

TABLE I
FEATURES EXTRACTED FROM SIGNAL

Feature Type	Feature Name and Number
Time domain Feature	F_1 : Root mean square; F_2 : Energy;
	F_3 : Entropy; F_4 : Max absolute;
Trigonometric Feature	F_5 : Mean absolute; F_6 : Standard deviation; F_7 : Clearance factor
	F_8 : standard deviation of inverse hyperbolic sine
Time-frequency domain Feature	$F_9 - F_{24}$: Energies of 16 bands
	TFRs

have clear physical meanings and there are numerous studies demonstrating that they reflect different state characteristics and degradation information of the bearings [18], [41]. It should be noted that stable features or indicators are almost meaningless for RUL prediction task. To enhance the robustness of the model and reduce the burden of learning, researchers have introduced a trend evaluation process, which is applied to select some of the best-trended time domain features [34], [42]. This process helps in reducing the dimensionality of input data and avoiding the interference of invalid information. Also, the energy of each frequency band processed by CWT is used as a time series feature since it has a good trend.

The features extracted from the original vibration signal are shown in Table I. The original features $F_1 - F_{24}$ are all vectors of length T , denoted as $F_i = [f_{i1}, \dots, f_{iT}]$, $i = 1, \dots, 24$, where T is the full life cycle length of the bearing. TFRs are tuples of length T , denoted as $\text{TFRs} = \{\text{TFR}_1, \dots, \text{TFR}_T\}$, where $\text{TFR}_i \in \mathbb{R}^{H \times W}$ ($i = 1, \dots, T$). From the whole life cycle data, the RUL value for each moment can be obtained and denoted as $\text{RUL} = [\text{rul}_1, \dots, \text{rul}_T]$. The dependencies between data at adjacent times are essential for the RUL prediction of bearing. Therefore, in order to consider the temporal dependence in both domains, a sliding time window (TW) is used as shown in Fig. 6. The width of TW is set to l , and the sliding step is 1. It is usually considered that the larger l is, the more information is contained in each TW. After the TW process, the data samples can be constructed as:

$$\begin{aligned}
 [X, Y] &= [(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)] \\
 &= \text{TW}(F_1, \dots, F_{24}, \text{TFRs}, \text{RUL}) \\
 &= [(f_{i1 \sim il}, \text{TFR}_{1 \sim l}, \text{rul}_l), \dots, \\
 &\quad (f_{in \sim iT}, \text{TFR}_{n \sim T}, \text{rul}_T)]
 \end{aligned} \tag{5}$$

where $[X, Y]$ is the training sample set, n is equal to $T - l + 1$, $i = 1, \dots, 24$, and $\text{TW}(\cdot)$ indicates TW processing.

B. CSA-DCNN: Feature Extraction Module for TFRs

The existing studies have concluded that the TFRs decoupled from the original vibration signal contain a large amount of useful degradation information [9], [20], [43]. In fact, TFRs contain not only degradation information, but also other state feature representations and noise. For example, some shocks caused by faults will exhibit a kind of energy aggregation on the TFRs after the time-frequency transformation, which is obviously the key degradation feature to be emphasized. Thus, identifying and

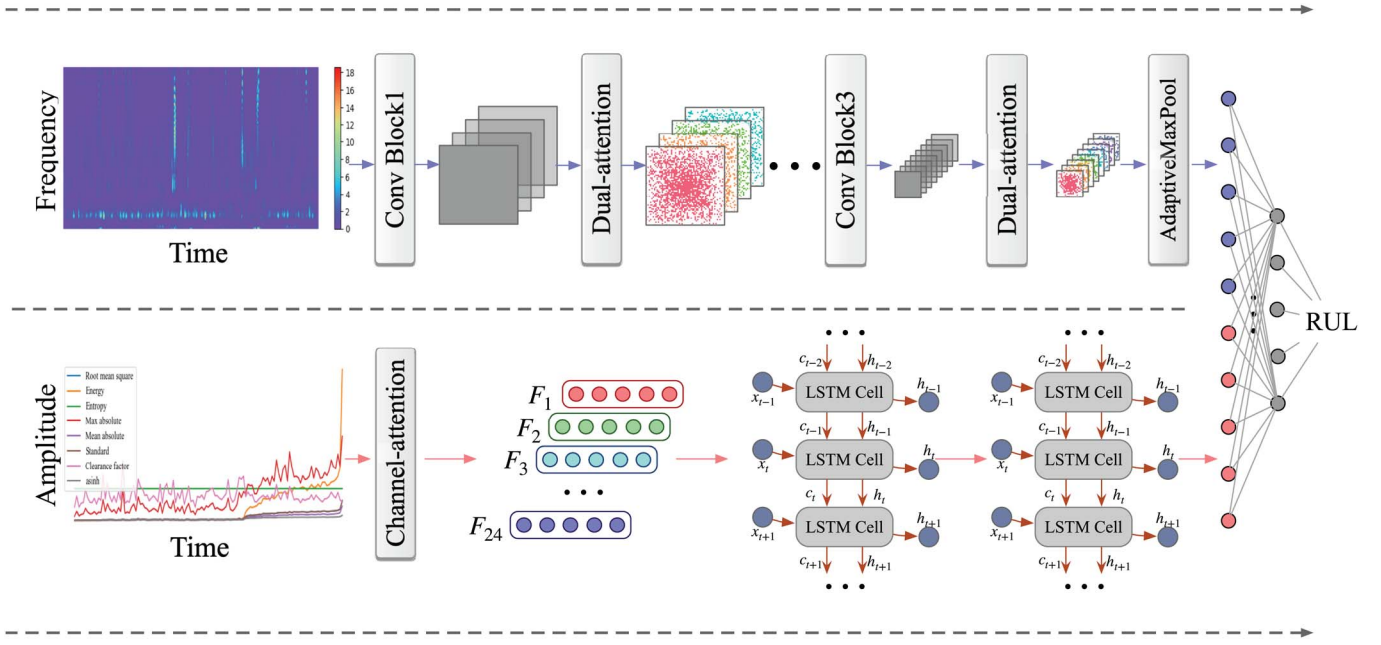


Fig. 4. Schematic diagram of the integrated workflow structure of DTADAN.

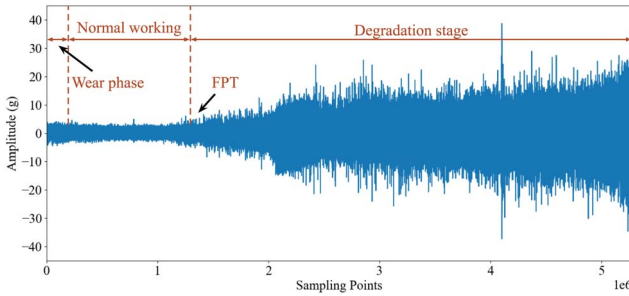


Fig. 5. FPT of raw bearing vibration signals.

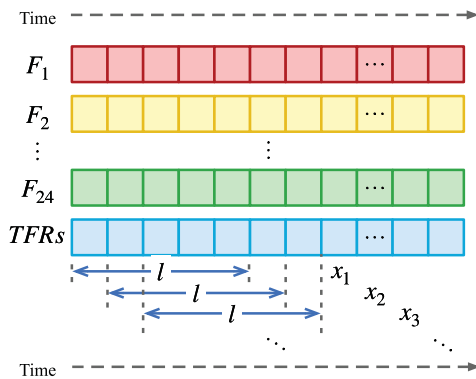


Fig. 6. Schematic diagram of TW processing.

extracting the typical features highly correlated with bearing deterioration plays a key role in improving prediction accuracy from the large amount of information on TFRs.

In addition, the degradation trend information in the time-frequency domain scale is also considered. By considering

the adjacent multistep history information through the TW operation, the receptive field is expanded to avoid data anomalies at a certain moment, and the trend information is supplemented by considering the nonlinear increase of the bearing degradation rate. Thus, degradation trend information is added to the RUL estimation as a piece of key degradation-related information [36].

The introduced dual attention considers the importance of the information extracted from TFRs in two dimensions. Embedding the dual attention into the DCNN structure can effectively enhance the extraction of useful information and suppress irrelevant information and noise. The CA mechanism enables the network structure to learn ‘what’ is effective, and the spatial attention mechanism detects ‘where’ is critical from it. Specifically, the DCNN performs multiple convolution operations on the raw input TFRs to extract features gradually from shallow to deep layers. In shallow convolutions, expanding the channel number extracts more information, while compressing it fuses different information in deep convolutions. Then, various channels capture features with differing amounts of information and correlation levels with degradation. Therefore, CA is applied to reweight these features, which helps extraction and fusion of important features in the next round. To capture key degradation information in these features, the position encoding generated by spatial attention is used to enhance critical degradation information. In particular, max pooling and average pooling are introduced to both types of attention. Max pooling filters out more useless information and highlights important features in the feature map, whereas average pooling tends toward overall characteristics to prevent too much high-dimensional loss of information. When extracting features related to degradation, average pooling extracts global information while max pooling focuses on prominent features. Combining these two types

of information can effectively capture degradation trends and improve structural perception ability comprehensively. With the two attention embeddings, CSA-DCNN obtains more powerful extraction and representation of degradation features.

C. CA-LSTM: Time Series Feature Fusion Module

Some DL methods have completed the task of RUL prediction by using the original vibration data as input and extracting the prediction-relevant features from it [44]. However, there are still two problems:

- 1) Feature extraction does not combine a lot of practical experience and physical knowledge. The extracted features are difficult to interpret the specific meaning.
- 2) Even if the network is simple, it still increases the computational cost due to the high dimensionality of the data.

The statistical indicators selected considering a large amount of a priori knowledge have a clear physical meaning, and these indicators have different meanings for describing the operating conditions of bearings. For example, the root mean square (RMS) and standard deviation (SD) indicate the dispersion, and the Clearance factor is often used to detect the wear of mechanical equipment [45], [46]. So their contribution to the RUL estimation varied at different moments. Some indicators may sometimes contain key information, while others do not contain useful information. Under this premise, it is obvious that the mining of degradation-related features will be hindered if these features are treated as equally important elements to be input into the LSTM.

In this article, eight time domain statistical indicators and 16 frequency band energies are considered as features of different channels. To mine the inherent connections between these pieces of information and adjust feature weights adaptively, CA blocks are utilized to process these indicators. Different indicators have varying sensitivities to bearing degradation at different stages. Although they all exhibit good trends, weight adjustment is still necessary for bearing degradation trend awareness based on the trend information. Introducing both max pooling and average pooling operations can simultaneously capture more stable information about changes in trends. Max pooling is highly sensitive to detailed degradation information but can be affected by outliers. Average pooling has stronger robustness but may overlook detailed degradation information. Combining these two pooling methods can enable CA to explore the potential connections between features, while also integrating information allocation with reasonable weights from both dimensions. Then the temporal information of time series data is considered through the LSTM structure. The combination of the two models enables the CA-LSTM model to learn the degraded state representation in the channel and time dimensions more effectively than before.

IV. EXPERIMENTAL CASE STUDY

Case Study I demonstrates that optimizing model structure can lead to more effective extraction of degradation trend information. This entails analyzing and refining the model structure

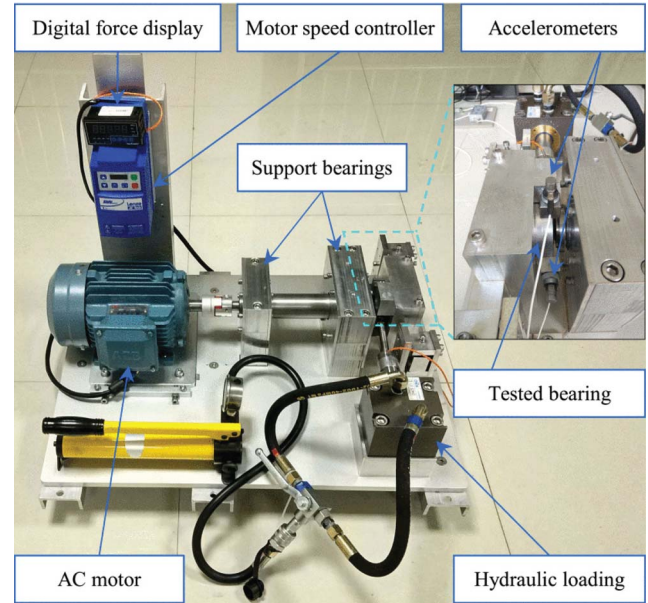


Fig. 7. Accelerated degradation platform for rolling element bearings.

to extract crucial features, such as degradation trend information, reduce interference from irrelevant information, and improve overall utilization of data.

In contrast, Case Study II takes a different approach by analyzing feature distributions and degradation trends to minimize interference from irrelevant information during model training. The goal is to highlight only the most effective information for accurate analysis and prediction.

PyTorch is utilized to implement all models and experiments. For further information on the experiments and basic code, please refer to the following GitHub repository: <https://github.com/CodeCollectorliu/DTADNN-2023.git>.

A. Case Study I

In order to fully demonstrate and evaluate the effectiveness of the proposed framework, a bearing experimental case XJTU-SY rolling bearing dataset [47] is applied in this article.

1) *Experimental Data Description:* This data set is comprised of vibration data for 15 rolling element bearings from accelerated degradation tests. The bearing test platform is shown in Fig. 7. In the accelerated degradation experiments of bearings, three subsets of bearings were operated at different speeds and loads until the end of life. The operating conditions include: 1) 12 kN/2100 rpm; 2) 11 kN/2250 rpm; and 3) 10 kN/2400 rpm. The detailed experimental bench operation setup can be found in [47].

Two accelerometers monitor the amplitude in horizontal and vertical directions, and the vibration signal in the horizontal direction is selected for RUL prediction in this article. Compared to the vertical vibration signal, the horizontal signal is less noisy, exhibits relatively stable changes, and performs better in displaying long-term trends. The sampling frequency is 25.6 kHz, each sample lasts 1.28 s, and the sampling interval is 1 min. so there will be 32 768 data points per sample. To

TABLE II
DETAILS OF THE XJTU-SY BEARING DATASET

Operating Condition	Radial Force/ Rotating Speed	Bearing Dataset	Bearing Lifetime	FPT
Condition 1	12 kN/ 2100 rpm	Bearing1_1	123 min	76 min
		Bearing1_2	161 min	42 min
		Bearing1_3	158 min	60 min
		Bearing1_4	122 min	-
		Bearing1_5	52 min	36 min
Condition 2	11 kN/ 2250 rpm	Bearing2_1	491 min	455 min
		Bearing2_2	161 min	48 min
		Bearing2_3	533 min	327 min
		Bearing2_4	42 min	30 min
		Bearing2_5	339 min	141 min
Condition 3	10 kN/ 2400 rpm	Bearing3_1	2538 min	2344 min
		Bearing3_2	2496 min	-
		Bearing3_3	371 min	340 min
		Bearing3_4	1515 min	1418 min
		Bearing3_5	114 min	7 min

reduce the data volume, 2560 sampling points are selected as one sampling data by referring to the practice in RONOSTIA bearing dataset from IEEE PHM 2012 data challenge [48]. The specific bearings and their FPT are detailed in Table II.

There are two bearings in the table with missing FPT because the FPT discrimination method is not applicable to the vibration signals of these two bearings. The failure modes of Bearing1_4 are sudden failures. Bearing3_2 has severe vibrations due to a mixture of multiple failure modes. Therefore, there are 13 bearings from the three conditions involved in the experiment.

2) *Experiment Preparation and Model Training*: The data between FPT and end-of-life (EoL) for 13 bearings are calculated and processed into time series features and TFRs according to the method proposed in Section III. However, the dimensionality of the obtained raw TFRs is high, so the bilinear interpolation method is used to reduce the dimensionality of the TFRs to 80×80 [20]. In this article, the time window length l is chosen as 5. Take Bearing 1_1 as an example, Input_1: Time series feature dimension is $43 \times 24 \times 5$, Input_2: TFRs dimension is $43 \times 80 \times 400$ and $X = \{\text{Input}_1, \text{Input}_2\}$. Y is the label indicating their RUL percentage (RULP) at each moment, which is constructed by the linear degradation model from FPT to EoL [43], [49].

When all multidirection sample sets $[X, Y]$ are constructed, the training set and test set are selected as follows. In each working condition, one bearing data is selected as the test set and the rest of the bearing data is used as the training set. In order to train the proposed DATDAN framework, the training set is randomly scrambled and input into the constructed network. The detailed parameter settings of the DTADAN are shown in Table III. The loss function used for model training is the MSELoss, and other hyperparameters can refer to Table IV. For testing, the corresponding X is obtained at each moment. Then X is input into the trained DTADAN parallel network, and the final output is RUL estimation.

3) *Evaluation and Comparison of Experimental Results*: To quantify the prediction performance of the proposed model for comparison with similar methods. In this article, two common

TABLE III
MAIN STRUCTURE OF DTADAN

Layer and Parameters Size	
TFRs direction	Time Series Direction
Batch_norm: $c = 5$	Channel_attention
Conv: $k = 10, c = 32, st = 5$	
Channel_attention	
Spatial_attention	
Batch_norm: $c = 32$	
Conv: $k = 5, c = 16, st = 2$	LSTM: out = 32
Channel_attention	
Spatial_attention	
Batch_norm: $c = 16$	
Conv: $k = 4, c = 8, st = 2$	
Channel_attention	Flatten
Spatial_attention	
Flatten	Flatten
Concat: 64	
FC: 1	

$c = 1$ means the number of channels is 1, etc.; $k = 5$ means the size of kernel is 5×5 , etc.; $st = 5$ means the size of stride is 5, etc.

TABLE IV
OTHER DETAILS AND HYPERPARAMETERS

Learning Rate	Dropout	Epoch	Batch	Optimizer
0.003	0.15	500	256	Adam

evaluation metrics, root mean square error (RMSE) in (6) and mean absolute error (MAE) in (7), are applied to evaluate the prediction results.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N_{\text{sample}}} (\hat{y}_i - y_i)^2}{N_{\text{sample}}}} \quad (6)$$

$$\text{MAE} = \sum_{i=1}^{N_{\text{sample}}} |\hat{y}_i - y_i| / N_{\text{sample}} \quad (7)$$

where N_{sample} is the number of test samples, $\hat{y}_i$ and y_i denote the predicted RULP value and the actual RULP value. The lower the RMSE and MAE of the method, the better performance of the method for RUL prediction.

To demonstrate the superiority of DTADAN, Tables V and VI show the results of comparing DTADAN with several advanced DL-based methods. Compared to other methods, DTADAN achieves impressive accuracy in all three operating conditions. Among them, MLP is a traditional and classical shallow ML method. In addition, BiLSTM models are good at handling time series features to capture the long time trend information, and have been effectively applied to prediction problems such as turbofan engines [51] and rolling bearings [50]. And MSCNN, as a multiscale DL method, has also been applied to the rolling bearing prediction method [9], [43], [52]. Finally, DCNN-MLP dual network is introduced as another hybrid prediction method for comparison with the proposed method [20]. It is important to note that the results provided in the table are based on 10

TABLE V
RMSE OF PREDICTION RESULTS FROM DIFFERENT ML MODELS

Models Test Bearing	DTADAN		DCNN-MLP [20]		MSCNN [9]		BiLSTM [50]		MLP	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Bearing1_1	0.206	0.023	0.206	0.025	0.242	0.022	0.228	0.020	0.274	0.016
Bearing1_2	0.208	0.025	0.240	0.023	0.262	0.028	0.305	0.013	0.313	0.029
Bearing1_3	0.103	0.017	0.178	0.049	0.184	0.040	0.130	0.015	0.261	0.029
Bearing1_5	0.138	0.054	0.184	0.066	0.215	0.063	0.362	0.029	0.318	0.027
Bearing2_1	0.088	0.009	0.117	0.028	0.148	0.053	0.152	0.016	0.203	0.046
Bearing2_2	0.102	0.013	0.122	0.031	0.232	0.084	0.134	0.012	0.266	0.038
Bearing2_3	0.158	0.008	0.158	0.047	0.199	0.052	0.216	0.031	0.230	0.020
Bearing2_4	0.162	0.029	0.177	0.031	0.231	0.054	0.311	0.032	0.251	0.033
Bearing2_5	0.092	0.016	0.092	0.022	0.108	0.029	0.308	0.022	0.234	0.050
Bearing3_1	0.121	0.036	0.244	0.073	0.247	0.068	0.351	0.030	0.305	0.059
Bearing3_3	0.133	0.034	0.158	0.026	0.191	0.029	0.188	0.015	0.318	0.047
Bearing3_4	0.197	0.014	0.132	0.021	0.165	0.026	0.175	0.019	0.252	0.036
Bearing3_5	0.239	0.042	0.266	0.060	0.267	0.049	0.305	0.036	0.376	0.042

The bold entries represent the best experimental evaluation results.

TABLE VI
MAE OF PREDICTION RESULTS FROM DIFFERENT ML MODELS

Models Test Bearing	DTADAN		DCNN-MLP [20]		MSCNN [9]		BiLSTM [50]		MLP	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Bearing1_1	0.175	0.025	0.176	0.018	0.213	0.015	0.191	0.017	0.240	0.013
Bearing1_2	0.166	0.024	0.207	0.016	0.229	0.014	0.231	0.013	0.270	0.022
Bearing1_3	0.085	0.015	0.151	0.042	0.155	0.041	0.106	0.012	0.221	0.025
Bearing1_5	0.105	0.041	0.155	0.057	0.181	0.054	0.314	0.025	0.265	0.021
Bearing2_1	0.068	0.007	0.099	0.029	0.126	0.051	0.129	0.020	0.172	0.036
Bearing2_2	0.078	0.010	0.102	0.027	0.194	0.074	0.094	0.009	0.214	0.037
Bearing2_3	0.123	0.008	0.126	0.037	0.164	0.045	0.170	0.025	0.204	0.018
Bearing2_4	0.127	0.024	0.141	0.024	0.195	0.046	0.267	0.029	0.213	0.023
Bearing2_5	0.073	0.016	0.075	0.200	0.090	0.026	0.278	0.022	0.202	0.044
Bearing3_1	0.098	0.031	0.204	0.062	0.214	0.057	0.297	0.027	0.262	0.049
Bearing3_3	0.109	0.027	0.129	0.025	0.156	0.023	0.162	0.012	0.276	0.030
Bearing3_4	0.170	0.012	0.107	0.020	0.139	0.017	0.135	0.017	0.220	0.028
Bearing3_5	0.193	0.037	0.219	0.046	0.225	0.038	0.251	0.031	0.310	0.036

The bold entries represent the best experimental evaluation results.

replicate trials to reduce the effect of randomness (μ and σ are the mean and SD of each bearing).

From an overall perspective, our proposed method performs better in terms of prediction accuracy than traditional DL models and other hybrid methods. It should be noted that the proposed methods in Bearing1_1, Bearing2_3, and Bearing3_4 do not improve tremendously. The reason why the proposed method fails to make much progress on Bearing1_1 will be given in the remaining two parts of the article. For Bearing2_3, although the RMSE score is close to that of the other hybrid method, the proposed method obtains a lower MAE score and has a significant improvement in stability (markedly reduced SD). The proposed method does not perform well on Bearing3_4, probably because the data characteristics of Bearing3_4 are more different from the remaining three training sets. When it is close to complete damage, it does not degrade as quickly as other bearings, but remains relatively stable for a period of time before complete damage, and then suddenly fails.

Some of the prediction results of the proposed model are shown in Fig. 8, from which it can be seen that the estimation error becomes smaller and smaller. And this is in line with the reality that as the degradation proceeds, the degradation information becomes richer and the trend characteristics become more and more obvious. The proposed model can capture

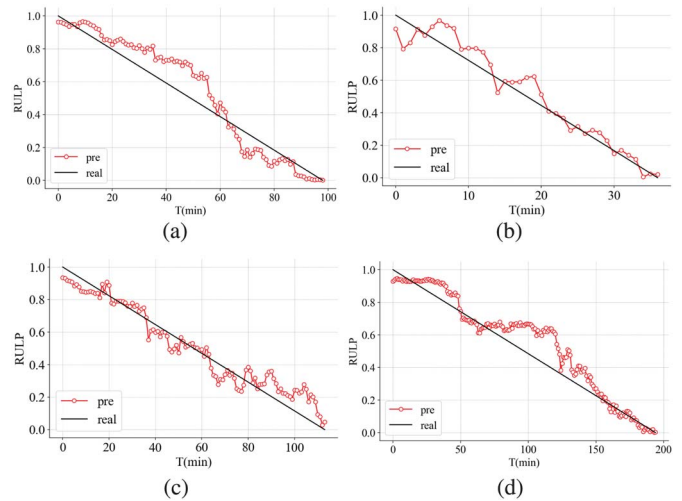


Fig. 8. RUL prediction results of the proposed method. (a) Bearing1_3. (b) Bearing2_1. (c) Bearing2_2. (d) Bearing3_1.

long-term trend information in both domains and key detail features closely related to degradation, thus maintaining a good trend without much fluctuation throughout the whole life cycle, and increasing prediction accuracy. This is very beneficial in practical industrial applications.

TABLE VII
RMSE OF PREDICTION RESULTS FROM DIFFERENT ATTENTION MODELS

Models Test Bearing	DTADAN		SARN [30]		SACLSTM [34]		DAMCNN [31]	
	μ	σ	μ	σ	μ	σ	μ	σ
Bearing1_1	0.206	0.023	0.213	0.034	0.269	0.016	0.197	0.033
Bearing1_2	0.208	0.025	0.212	0.028	0.221	0.020	0.241	0.039
Bearing1_3	0.103	0.017	0.136	0.016	0.156	0.033	0.206	0.034
Bearing1_5	0.138	0.054	0.169	0.021	0.164	0.024	0.224	0.052
Avg	0.164	0.029	0.182	0.025	0.180	0.023	0.217	0.039

The bold entries represent the best experimental evaluation results.

TABLE VIII
MAE OF PREDICTION RESULTS FROM DIFFERENT ATTENTION MODELS

Models Test Bearing	DTADAN		SARN [30]		SACLSTM [34]		DAMCNN [31]	
	μ	σ	μ	σ	μ	σ	μ	σ
Bearing1_1	0.175	0.025	0.170	0.025	0.206	0.013	0.161	0.028
Bearing1_2	0.166	0.024	0.171	0.025	0.179	0.021	0.187	0.031
Bearing1_3	0.085	0.015	0.113	0.016	0.133	0.027	0.172	0.032
Bearing1_5	0.105	0.041	0.140	0.015	0.151	0.064	0.181	0.033
Avg	0.133	0.026	0.149	0.020	0.168	0.032	0.175	0.033

The bold entries represent the best experimental evaluation results.

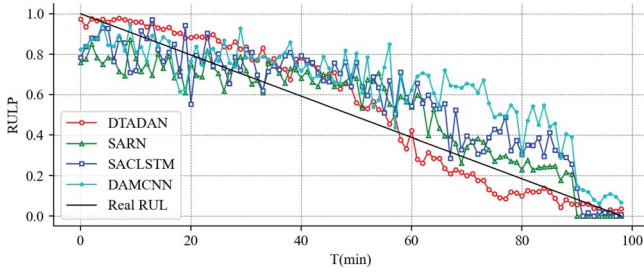


Fig. 9. Comparison of different methods with attention on Bearing1_3.

In addition, the proposed framework still has advantages over some existing typical attention-based approaches. Taking Bearings1_3 as an example in Fig. 9, the proposed method is closer to the true value compared to other attention methods and tends to give conservative RUL estimation when approaching the end of bearing life, which is critical for equipment operation safety. The constructed attention structure can achieve good degradation trend awareness, especially at the end of degradation. In Tables VII and VIII, the rich information fusion of residual structures in SARN [30] makes it perform well in degraded feature extraction. However, SARN [30] only adjusts attention for deep features, which greatly limits further enhancement of predictive ability. Although the self-attention mechanism improves the prediction performance of SACLSTM [34] significantly, it still has significant fundamental flaws. Its degradation feature extraction capability is weaker than that of others, and weight adjustment between hidden states and inputs with self-attention is difficult to interpret. DAMCNN [31] introduces dual attention and recurrent structures to improve prediction accuracy compared to MSCNN, but only applying attention to raw input is unsuitable for complex vibration signals with low signal-to-noise ratio.

4) *Effect of Time-Dependent Trend Information*: It is quite evident that degradation trend perception of time series type features is easy to achieve because they are selected based on

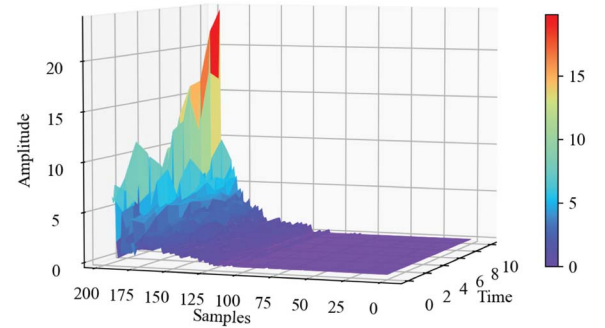


Fig. 10. Visualization of features extracted from TFRs.

TABLE IX
ENHANCEMENT OF RMSE AND MAE ON DIFFERENT MODELS

Models Test Bearing	DTADN		DCNN-MLP [20]	
	RMSE	MAE	RMSE	MAE
Bearing1_1	0.207	0.173	0.206	0.176
Bearing1_3	0.125	0.104	0.178	0.151
Bearing2_1	0.111	0.088	0.117	0.099
Bearing3_1	0.233	0.189	0.244	0.219

The bold entries represent the best experimental evaluation results.

trending metrics. However, the proposed method also learns the degradation trend information from TFRs. As shown in Fig. 10, where “sample” denotes sample points from healthy to failure in life cycle and “time” refers to the occurrence order of the characteristics of each sample over time. They also show an increasing trend as the degradation proceeds. Therefore, it is necessary to consider the information fusion of degradation trends for different direction representations.

In this subsection, we will examine the importance of the degradation trend information. For the sake of notation, DTADN can be regarded as the remaining structure after removing all attention blocks from the structure shown in Fig. 4. In Table IX, neither DTADN nor the DCNN-MLP structure

TABLE X
ENHANCEMENT OF RMSE AND MAE ON DIFFERENT MODELS

Models Test Bearing	DTADAN		DTADCAN		DTADCSAN		DTADN	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Bearing1_1	0.206	0.175	0.206	0.174	0.206	0.179	0.207	0.173
Bearing1_3	0.103	0.085	0.118	0.093	0.116	0.094	0.125	0.104
Bearing2_1	0.088	0.068	0.103	0.083	0.095	0.074	0.111	0.088
Bearing3_1	0.121	0.098	0.205	0.162	0.143	0.112	0.233	0.189

The bold entries represent the best experimental evaluation results.

contains attention mechanism, the degradation trend feature is captured by TW processing and LSTM structure in the experiments of DTADN. The structure of DTADN can be referred to the remaining structure after removing all attention blocks from the structure shown in Fig. 4. The comparison results show that the introduction of degradation trends in both submodules can additionally complement the trend features associated with the degradation process, which is more beneficial for RUL prediction than considering only the features at each moment.

With the introduction of the TW, more information is available for the RUL prediction model. Because bearing damage is continuous and nonlinear, the trends of degradation-related features vary at different stages. It is obvious that such information cannot be captured if only the features at a certain moment are utilized. In addition, direct input to the network for some data outliers will give wrong prediction results. In contrast, the prediction method will be more robust and noise-resistant due to the utilization of the surrounding data features.

It should be added that the change of the structure of Bearing1_1 does not improve the accuracy, because the data of Bearing1_1 are relatively pure and the degradation process tends to be linear, so the simple structure can achieve a good prediction accuracy. However, for some more complex situations, the accuracy and robustness of RUL prediction are significantly improved.

5) *Effect of Attention Structure:* However, since DTADN does not introduce attention mechanism at this time, although the accuracy is improved, it is not obvious in general. This is due to the fact that although the input multidirection features make the information richer, the difficulty of feature extraction and utilization also increases, and DTADN is clearly weak in feature discrimination. Therefore, to better distinguish between various features and streamline the information selection process, the attention design in our proposed framework takes into account the unique characteristics of vibration data and specific raw inputs. This approach highlights degradation-related information while minimizing noise and irrelevant variables, resulting in improved degradation trend awareness of the network.

To demonstrate the effectiveness of the attention module, DTADCSAN and DTADCAN are constructed for comparison experiments, as shown in Table X. Among them, DTADCSAN and DTADCAN are two models that incorporate attention mechanisms in different ways. DTADCSAN adds both channel and spatial attention mechanisms to the TFR processing branch, while DTADCAN only includes a CA mechanism in the time series processing branch. The structure of DTADCSAN is based on Fig. 4, but removes all CA blocks while keeping the dual attention block. In contrast, the structure of DTADCAN

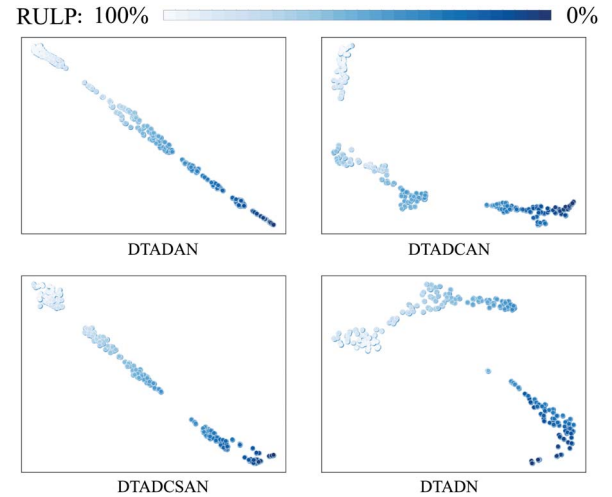


Fig. 11. Features visualization by T-sne for different models on Bearing3_1.

retains all CA blocks but eliminates all dual attention blocks. Compared with DTADN, both DTADCSAN and DTADCAN improve the accuracy of RUL prediction to varying degrees, which indicates that both parts of the attention module improve the feature extraction ability of the framework. The partial removal of attention modules results in information loss. These two models have incomplete degradation information in the high-level features in comparison with DTADAN. The loss of degradation trend and detailed information will affect the final prediction accuracy. Thus, incorporating attention into these parts is crucial for improving RUL prediction performance.

Then, as shown in Fig. 11, the four models are visualized on Bearing 3_1 by T-sne. The features extracted by DTADAN and DTADCSAN have a very clear linear trend. And due to the inclusion of the two attention models, DTADAN is clearly more densely and regularly distributed than DTADCSAN, especially near the EoL. The accuracy of DTADCAN is relatively a bit worse. It can also be seen from its feature distributions that it has a certain distribution regularity in general, showing nonlinearity, but the degree of dispersion is relatively high. The accuracy of DTADN is the worst, and it can be found from the figure that the feature distributions are scattered and has almost no trend.

Additionally, it can be noticed that the prediction performance on Bearing1_1 shows no significant difference. This is because the data of Bearing1_1 is relatively pure and degradation features are relatively easy to distinguish, so the deep convolution operation can readily capture the corresponding degradation features. However, for more complex signals, the

TABLE XI
DETAILS OF THE PRONOSTIA BEARING DATASET

Operating Condition	Radial Force/ Rotating Speed	Bearing Dataset	Bearing Lifetime	FPT
Condition 1	4 kN/ 1800 rpm	Bearing1_1	2803 (10s)	1462 (10s)
		Bearing1_2	871 (10s)	826 (10s)
		Bearing1_3	2375 (10s)	1726 (10s)
		Bearing1_4	1428 (10s)	1082 (10s)
		Bearing1_5	2463 (10s)	2412 (10s)
		Bearing1_6	2448 (10s)	2394 (10s)
		Bearing1_7	2259 (10s)	2210 (10s)

introduction of the attention mechanism leads to a better feature extraction capability for the proposed method.

B. Case Study II

In order to demonstrate the generalization ability of the proposed method, we introduced another more complex bearing dataset, PRONOSTIA bearing dataset from IEEE PHM 2012 data challenge [48]. Additionally, in this subsection, a universal concept of feature distributions and degradation trends will be demonstrated.

Considering the different failure modes and individual variability of bearings, the degradation patterns of bearings also vary. Based on this universal concept, an innovative optimization strategy is presented to effectively address the differences among distinct degradation patterns, which enables the DL model to achieve higher accuracy and robustness with smaller computational costs during training.

1) *Experimental Data Description*: This dataset is a publicly available rolling bearing dataset, and the relevant experimental platform parameters can be found in [48]. In this experiment, 7 complete lifecycle bearings in the first operating state as shown in Table XI were used. These bearings exhibited different degradation patterns.

2) *Experiment Preparation and Model Training*: In order to demonstrate the general applicability of the proposed method and compare different methods fairly. The same preprocessing methods and model parameters as the previous case were used in this experimental case. In addition, construction of samples and RULP labels were also completely identical.

The model training strategy will be divided into three steps to better utilize feature distributions and degradation trends to cope with the problem of diverse degradation patterns.

- 1) To begin with, all bearings except those used for testing were utilized as a training set to incorporate more degraded pattern samples. It is necessary to include some healthy samples labeled with one in the model training process. This round of DATDNN training will serve as a benchmark.
- 2) Then, the data collected from monitoring the test bearing's early degradation is used to create samples. Then these samples are inputted into the benchmark to obtain the feature distributions and degradation trend of the initial deterioration stage of the tested bearings and those in the known training dataset.
- 3) Finally, two bearings were chosen from the known data as an optimized training set based on the similarity of their

feature distributions and degradation trends. The model trained using this set was referred to as DATDNN-O. Subsequently, the RUL prediction results for the tested bearing were obtained by DATDNN-O.

3) *Visualization of Features and Principle Analysis*: In complex datasets, the degradation patterns of bearings are often not singular. Taking this dataset as an example, it contains a slow degradation pattern, such as Bearing1_1, a short-term rapid degradation pattern, such as Bearing1_6, and data that simultaneously exhibit both types of degradation patterns, such as Bearing1_3. The feature distributions and degradation trends of various degradation patterns differ significantly. As a result, using complex and chaotic data for DL training without considering these differences can increase training costs and hinder achieving the desired accuracy. To address these issues, it is crucial to match the distribution and degradation trends appropriately.

The proposed optimization strategy aims to select known training bearings that closely approximate the feature distributions and degradation trends of the test bearing. Initially, a benchmark is trained using all available bearings to capture and characterize the feature distributions and degradation trends of the training bearings effectively. For this reason, in the case of monitoring the initial degradation state, this benchmark can be utilized to offer an estimated distribution pattern and trend of deterioration for the bearings under test. The feature distributions and degradation trends were derived from benchmark tests. Therefore, choosing training bearings with comparable performance that are already known is both precise and dependable.

Subsequently, the method optimized with highly similar bearings has clear advantages and can effectively extract degradation trend information from future monitoring data, leading to a more accurate RUL prediction. Even though predicting the degradation pattern of the test bearing is challenging, and even the degradation patterns of bearings in the known training set are unknown, prematching feature distributions and degradation trends under benchmark can overcome this issue during DL model training. This strategy significantly improves RUL prediction accuracy at a lower cost in complex situations.

This optimization strategy is straightforward and aligns with human intuition. In Fig. 12, Bearing1_1 and Bearing1_6 are used as two typical examples of test bearings, to demonstrate their feature distributions and degradation trends in comparison to the other six training bearings under benchmark (left image). Additionally, it showcases only retaining the feature distributions and trends of training bearings that exhibit high similarity (right image). Note that color fading indicates a gradual increase in damage severity, while different colors are used to distinguish between various bearings.

Based on the proposed strategy, Bearing1_3 and Bearing1_7 are recommended as the most suitable matches for feature distributions and degradation trends when compared to Bearing1_1. Similarly, to optimize the performance of Bearing1_6, it is recommended to use Bearing1_2 and Bearing1_7 as training sets because their feature distributions are more similar. However, there is a significant difference in degradation trend

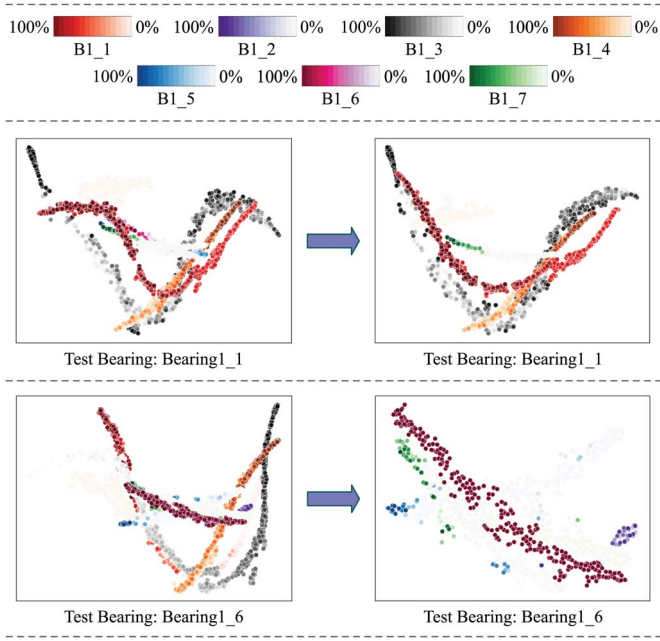


Fig. 12. Degradation trends and distribution of bearing features.

TABLE XII
TRAINING SET COMPOSITION FOR DATDNN-O

Test Bearing	Train Bearings
Bearing1_1	Bearing1_3, Bearing1_4
Bearing1_2	Bearing1_6, Bearing1_7
Bearing1_3	Bearing1_1, Bearing1_7
Bearing1_4	Bearing1_3, Bearing1_7
Bearing1_5	Bearing1_3, Bearing1_7
Bearing1_6	Bearing1_2, Bearing1_7
Bearing1_7	Bearing1_1, Bearing1_5

of Bearing1_5. For readers' convenience, Table XII provides training sets for all bearings.

4) *Evaluation and Comparison of Experimental Results:* In this case study, RMSE and MAE are still used as evaluation metrics to quantify predictive performance for comparison.

In this subsection, DTADAN-O, regular DTADAN and DCNN-MLP with bootstrap [20] were used for RUL prediction and compared. All experiments were repeated 10 times to calculate the mean and SD. The RMSE score results are shown in Table XIII, and the MAE score results are shown in Table XIV.

In the tables, DTADAN-O is the optimized version of DATDNN using our proposed strategy. DTADAN represents the unoptimized method. DCNN-MLP denotes a hybrid method with bootstrap that uses three bearings as training set [20]. The results in the table demonstrate that our proposed method has good prediction performance on this dataset. Furthermore, optimizing with our strategy consistently improves both prediction accuracy and robustness.

For instance, when considering Bearing1_6, the prediction performance may be inadequate if optimization strategies are not employed. This is because other methods tend

TABLE XIII
RMSE OF PREDICTION RESULTS FROM OPTIMIZED AND UNOPTIMIZED MODELS

Models Test Bearing	DTADAN-O		DTADAN		DCNN-MLP	
	μ	σ	μ	σ	μ	σ
Bearing1_1	0.138	0.023	0.158	0.034	0.194	0.024
Bearing1_2	0.155	0.047	0.200	0.043	0.254	0.025
Bearing1_3	0.151	0.011	0.159	0.008	0.199	0.009
Bearing1_4	0.118	0.023	0.128	0.026	0.132	0.026
Bearing1_5	0.163	0.020	0.186	0.018	0.187	0.024
Bearing1_6	0.183	0.016	0.293	0.038	0.328	0.007
Bearing1_7	0.200	0.042	0.198	0.041	0.205	0.029

The bold entries represent the best experimental evaluation results.

TABLE XIV
MAE OF PREDICTION RESULTS FROM OPTIMIZED AND UNOPTIMIZED MODELS

Models Test Bearing	DTADAN-O		DTADAN		DCNN-MLP	
	μ	σ	μ	σ	μ	σ
Bearing1_1	0.108	0.019	0.123	0.027	0.161	0.021
Bearing1_2	0.123	0.035	0.153	0.036	0.219	0.023
Bearing1_3	0.111	0.008	0.112	0.006	0.164	0.009
Bearing1_4	0.094	0.018	0.100	0.020	0.107	0.022
Bearing1_5	0.125	0.014	0.147	0.011	0.158	0.022
Bearing1_6	0.151	0.017	0.228	0.045	0.270	0.008
Bearing1_7	0.163	0.038	0.160	0.037	0.172	0.029

The bold entries represent the best experimental evaluation results.

to be influenced by excessive interference information in the training set. However, adopting optimization strategy can effectively eliminate invalid samples and enhance the model's adaptability, leading to a significant improvement in prediction accuracy.

In addition, we can find that there is no significant enhancement in the prediction accuracy of Bearing1_7. This can be attributed to the fact that, apart from Bearing1_5 which exhibits a comparable distribution, none of the other bearings share similar feature distributions and degradation trends with Bearing1_7. Consequently, it is currently unfeasible to identify a specific training set for targeted optimization. It follows logically that models trained on limited samples will perform inferiorly on unknown data compared to those trained on a larger dataset. Adequate availability of known data can prevent such situations from arising.

V. CONCLUSION AND FUTURE WORK

This article proposes a DTA RUL prediction framework with interpretability analysis of the prediction model. The multidirection feature data is composed of statistical indicators and TFRs with clear physical meanings. After being processed by TW, they serve as raw inputs containing rich state information and irrelevant information. In order to effectively mine the degradation information related to RUL prediction in different direction representations, different DTA structures are adopted to capture key degradation characteristics related to RUL prediction. The attention mechanism introduced in CSA-DCNN and CA-LSTM can improve the discriminative and

noise suppression ability of the model during feature extraction in each direction. It can also reduce learning difficulties caused by high-dimension and complexity resulting from multidirection representations.

Subsequently, a latent space is obtained by fusing key features, which can help the model train and achieve interpretable RUL prediction for unknown bearing data according to the proposed optimization strategy. The framework is applied to XJTU-SY rolling bearing dataset and PRONOSTIA bearing dataset, and the experimental results show excellent performance. The proposed method conducts in-depth interpretability research and visual analysis of data, degradation trends, attention mechanisms, and optimization strategy. Compared to traditional RUL prediction methods, this framework provides a better understanding of data structure, model learning process, and reasons behind enhanced prediction accuracy.

In further research, how to combine the proposed method with transfer learning and the interpretability analysis of the model learning at this stage are both worth considering. It is worthwhile to consider conducting RUL prediction under different operating conditions, which is more in line with engineering practice and can further improve the practical application value of the model.

AUTHORS' CONTRIBUTIONS

Yongkang Liu performed Visualization, Software, Methodology, and Writing - Original Draft; Donghui Pan performed Conceptualization, Investigation, Validation, and Writing - Review and Editing; Haifeng Zhang performed Supervision, Review, and Funding Acquisition; Kai Zhong performed Review, Validation, and Funding Acquisition.

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